anomaly_detection_using_clustering_algorithms

April 20, 2026

1 Anomaly Detection using Clustering Algorithms

This notebook performs anomaly detection on network-intrusion data using three unsupervised clustering algorithms:

- KMeans
- Agglomerative Clustering
- DBSCAN

The data is taken from the existing `AnamolyDetection/data` folder. The `class` label is **not used for clustering**; it is used only after clustering to evaluate how well each algorithm separated normal and anomalous traffic.

1.1 1. Imports and Settings

[1]:
```python
import os
from pathlib import Path
import warnings

os.environ.setdefault("MPLCONFIGDIR", "/tmp")
warnings.filterwarnings("ignore", category=FutureWarning)

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
import seaborn as sns
from IPython.display import display
from scipy.cluster.hierarchy import dendrogram, linkage

from sklearn.cluster import AgglomerativeClustering, DBSCAN, KMeans
from sklearn.decomposition import PCA
from sklearn.metrics import (
    calinski_harabasz_score,
    ConfusionMatrixDisplay,
    accuracy_score,
    classification_report,
    confusion_matrix,
    davies_bouldin_score,
    f1_score,
```

```
    precision_recall_fscore_support,
    silhouette_score,
)
from sklearn.neighbors import NearestNeighbors
from sklearn.preprocessing import StandardScaler

RANDOM_STATE = 42
SAMPLE_PER_CLASS = 1500
BW_HATCHES = ["///", "\\\\\\", "xx", "..", "++", "oo"]
BW_COLORS = ["white", "#E6E6E6", "#BFBFBF", "#8C8C8C", "#666666", "#D9D9D9"]
BW_MARKERS = ["o", "s", "^", "D", "x", "P", "*"]
BW_LINESTYLES = ["-", "--", "-.", ":"]

sns.set_theme(style="whitegrid")
plt.rcParams.update({
    "figure.dpi": 120,
    "axes.edgecolor": "black",
    "axes.labelcolor": "black",
    "xtick.color": "black",
    "ytick.color": "black",
    "text.color": "black",
})
pd.set_option("display.max_columns", 120)
pd.set_option("display.max_colwidth", 120)
```

1.2 1.1 Helper Functions

[2]:
```
def add_bar_labels(ax, horizontal=False):
    for patch in ax.patches:
        value = patch.get_width() if horizontal else patch.get_height()
        if np.isnan(value):
            continue
        if horizontal:
            ax.annotate(
                f"{value:,.0f}",
                (value, patch.get_y() + patch.get_height() / 2),
                ha="left",
                va="center",
                xytext=(4, 0),
                textcoords="offset points",
                fontsize=9,
            )
        else:
            ax.annotate(
                f"{value:,.0f}",
                (patch.get_x() + patch.get_width() / 2, value),
                ha="center",
```

```
                va="bottom",
                xytext=(0, 3),
                textcoords="offset points",
                fontsize=9,
            )


def plot_count_bar(counts, title, xlabel="", ylabel="", horizontal=False,␣
↪top_n=None):
    plot_counts = counts.copy()
    if top_n is not None:
        plot_counts = plot_counts.head(top_n)
    fig, ax = plt.subplots(figsize=(9, max(4, 0.35 * len(plot_counts))) if␣
↪horizontal else (8, 4.5))
    colors = [BW_COLORS[i % len(BW_COLORS)] for i in range(len(plot_counts))]
    if horizontal:
        bars = ax.barh(plot_counts.index.astype(str), plot_counts.values,␣
↪color=colors, edgecolor="black", linewidth=1.0)
        ax.invert_yaxis()
        ax.set_xlabel(xlabel)
        ax.set_ylabel(ylabel)
        ax.margins(x=0.15)
    else:
        bars = ax.bar(plot_counts.index.astype(str), plot_counts.values,␣
↪color=colors, edgecolor="black", linewidth=1.0)
        ax.set_xlabel(xlabel)
        ax.set_ylabel(ylabel)
        ax.tick_params(axis="x", rotation=25)
        ax.margins(y=0.15)
    for idx, bar in enumerate(bars):
        bar.set_hatch(BW_HATCHES[idx % len(BW_HATCHES)])
    ax.set_title(title)
    add_bar_labels(ax, horizontal=horizontal)
    plt.tight_layout()
    plt.show()


def get_data_root():
    candidates = [
        Path("AnamolyDetection/data"),
        Path("../data"),
        Path.cwd() / "AnamolyDetection" / "data",
    ]
    return next(path for path in candidates if path.exists())


def stratified_sample(data, label_col, n_per_class=SAMPLE_PER_CLASS):
```

```python
    parts = []
    for label_value, group in data.groupby(label_col):
        parts.append(group.sample(min(len(group), n_per_class), random_state=RANDOM_STATE))
    return pd.concat(parts).sample(frac=1, random_state=RANDOM_STATE).reset_index(drop=True)


def preprocess_features(data, label_col):
    feature_data = data.drop(columns=[label_col]).copy()
    categorical_cols = feature_data.select_dtypes(include=["object", "string", "category"]).columns.tolist()
    processed = pd.get_dummies(feature_data, columns=categorical_cols, drop_first=False)
    processed = processed.apply(pd.to_numeric, errors="coerce").replace([np.inf, -np.inf], np.nan).fillna(0)
    scaler = StandardScaler()
    scaled = scaler.fit_transform(processed)
    return processed, scaled, scaler


def map_clusters_to_anomaly(cluster_labels, y_true):
    mapped = np.zeros_like(y_true, dtype=int)
    mapping_rows = []
    for cluster_id in sorted(pd.Series(cluster_labels).unique()):
        mask = cluster_labels == cluster_id
        if cluster_id == -1:
            mapped_label = 1
            rule = "DBSCAN noise -> anomaly"
        else:
            counts = pd.Series(y_true[mask]).value_counts()
            mapped_label = int(counts.idxmax())
            rule = "majority label for evaluation"
        mapped[mask] = mapped_label
        mapping_rows.append(
            {
                "cluster": cluster_id,
                "mapped_prediction": "anomaly" if mapped_label == 1 else "normal",
                "cluster_size": int(mask.sum()),
                "true_anomaly_rate": float(y_true[mask].mean()) if mask.sum() else np.nan,
                "rule": rule,
            }
        )
    return mapped, pd.DataFrame(mapping_rows)
```

```python
def internal_cluster_metrics(X, cluster_labels, ignore_noise=True):
    labels = np.asarray(cluster_labels)
    if ignore_noise:
        valid_mask = labels != -1
        X_eval = X[valid_mask]
        labels_eval = labels[valid_mask]
    else:
        X_eval = X
        labels_eval = labels

    unique_labels = np.unique(labels_eval)
    if X_eval.shape[0] < 2 or len(unique_labels) < 2:
        return {
            "Silhouette Score": np.nan,
            "Davies-Bouldin Index": np.nan,
            "Calinski-Harabasz Index": np.nan,
            "Metric Notes": "Not enough non-noise clusters",
        }

    return {
        "Silhouette Score": silhouette_score(X_eval, labels_eval),
        "Davies-Bouldin Index": davies_bouldin_score(X_eval, labels_eval),
        "Calinski-Harabasz Index": calinski_harabasz_score(X_eval, labels_eval),
        "Metric Notes": "Noise ignored" if ignore_noise and np.any(labels == 
 ↪-1) else "All points used",
    }


def evaluate_clustering(name, cluster_labels, y_true, X_for_internal_metrics):
    y_pred, mapping_df = map_clusters_to_anomaly(cluster_labels, y_true)
    precision, recall, f1, _ = precision_recall_fscore_support(
        y_true,
        y_pred,
        average="weighted",
        zero_division=0,
    )
    result = {
        "Model": name,
        "Clusters": len(set(cluster_labels)) - (1 if -1 in cluster_labels else 
 ↪0),
        "Noise Points": int((cluster_labels == -1).sum()),
        "Accuracy": accuracy_score(y_true, y_pred),
        "Weighted Precision": precision,
        "Weighted Recall": recall,
        "Weighted F1": f1,
```

```
        "Anomaly F1": f1_score(y_true, y_pred, zero_division=0),
    }
    result.update(internal_cluster_metrics(X_for_internal_metrics,␣
↪cluster_labels, ignore_noise=True))

    print(f"{name} cluster-to-label mapping")
    display(mapping_df)

    print(f"{name} classification report")
    display(pd.DataFrame(classification_report(y_true, y_pred,␣
↪target_names=["normal", "anomaly"], output_dict=True, zero_division=0)).T.
↪round(4))

    cm = confusion_matrix(y_true, y_pred, labels=[0, 1])
    disp = ConfusionMatrixDisplay(confusion_matrix=cm,␣
↪display_labels=["normal", "anomaly"])
    disp.plot(cmap="Greys", values_format="d", colorbar=False)
    plt.title(f"{name} Confusion Matrix")
    plt.tight_layout()
    plt.show()
    return result, y_pred


def plot_pca_scatter(pca_df, color_col, title):
    fig, ax = plt.subplots(figsize=(8, 5.5))
    labels = sorted(pca_df[color_col].unique(), key=lambda x: str(x))
    for idx, label_value in enumerate(labels):
        subset = pca_df[pca_df[color_col] == label_value]
        ax.scatter(
            subset["PC1"],
            subset["PC2"],
            s=18,
            facecolors="none" if idx % 2 == 0 else BW_COLORS[idx %␣
↪len(BW_COLORS)],
            edgecolors="black",
            marker=BW_MARKERS[idx % len(BW_MARKERS)],
            linewidths=0.7,
            alpha=0.75,
            label=str(label_value),
        )
    ax.set_title(title)
    ax.set_xlabel("PC1")
    ax.set_ylabel("PC2")
    ax.legend(title=color_col, bbox_to_anchor=(1.02, 1), loc="upper left")
    plt.tight_layout()
    plt.show()
```

```python
def plot_metric_lines(df, x_col, y_cols, title, ylabel):
    fig, ax = plt.subplots(figsize=(8.5, 4.8))
    for idx, col in enumerate(y_cols):
        ax.plot(
            df[x_col],
            df[col],
            color=BW_COLORS[min(idx + 1, len(BW_COLORS) - 1)] if idx else␣
↪"black",
            marker=BW_MARKERS[idx % len(BW_MARKERS)],
            linestyle=BW_LINESTYLES[idx % len(BW_LINESTYLES)],
            linewidth=1.8,
            label=col,
        )
    ax.set_title(title)
    ax.set_xlabel(x_col)
    ax.set_ylabel(ylabel)
    ax.legend()
    plt.tight_layout()
    plt.show()
```

1.3 2. Load Dataset

[3]:
```python
DATA_ROOT = get_data_root()
raw_path = DATA_ROOT / "raw" / "Network_Intrusion_Detection" / "Train_data.csv"

df_raw = pd.read_csv(raw_path)
print("Using dataset:", raw_path.resolve())
print("Raw shape:", df_raw.shape)
display(df_raw.head())
```

```
Using dataset: /home/durgaumadev/security-for-data-science-
lab/AnamolyDetection/data/raw/Network_Intrusion_Detection/Train_data.csv
Raw shape: (25192, 42)
```

```
   duration protocol_type   service flag  src_bytes  dst_bytes  land  \
0         0           tcp  ftp_data   SF        491          0     0
1         0           udp     other   SF        146          0     0
2         0           tcp   private   S0          0          0     0
3         0           tcp      http   SF        232       8153     0
4         0           tcp      http   SF        199        420     0

   wrong_fragment  urgent  hot  num_failed_logins  logged_in  num_compromised  \
0               0       0    0                  0          0                0
1               0       0    0                  0          0                0
2               0       0    0                  0          0                0
3               0       0    0                  0          1                0
4               0       0    0                  0          1                0
```

```
   root_shell  su_attempted  num_root  num_file_creations  num_shells  \
0           0             0         0                   0           0
1           0             0         0                   0           0
2           0             0         0                   0           0
3           0             0         0                   0           0
4           0             0         0                   0           0

   num_access_files  num_outbound_cmds  is_host_login  is_guest_login  count  \
0                 0                  0              0               0      2
1                 0                  0              0               0     13
2                 0                  0              0               0    123
3                 0                  0              0               0      5
4                 0                  0              0               0     30

   srv_count  serror_rate  srv_serror_rate  rerror_rate  srv_rerror_rate  \
0          2          0.0              0.0          0.0              0.0
1          1          0.0              0.0          0.0              0.0
2          6          1.0              1.0          0.0              0.0
3          5          0.2              0.2          0.0              0.0
4         32          0.0              0.0          0.0              0.0

   same_srv_rate  diff_srv_rate  srv_diff_host_rate  dst_host_count  \
0           1.00           0.00                0.00             150
1           0.08           0.15                0.00             255
2           0.05           0.07                0.00             255
3           1.00           0.00                0.00              30
4           1.00           0.00                0.09             255

   dst_host_srv_count  dst_host_same_srv_rate  dst_host_diff_srv_rate  \
0                  25                    0.17                    0.03
1                   1                    0.00                    0.60
2                  26                    0.10                    0.05
3                 255                    1.00                    0.00
4                 255                    1.00                    0.00

   dst_host_same_src_port_rate  dst_host_srv_diff_host_rate  \
0                         0.17                         0.00
1                         0.88                         0.00
2                         0.00                         0.00
3                         0.03                         0.04
4                         0.00                         0.00

   dst_host_serror_rate  dst_host_srv_serror_rate  dst_host_rerror_rate  \
0                  0.00                      0.00                  0.05
1                  0.00                      0.00                  0.00
2                  1.00                      1.00                  0.00
3                  0.03                      0.01                  0.00
```

```
4                   0.00                      0.00                0.00

   dst_host_srv_rerror_rate     class
0                      0.00    normal
1                      0.00    normal
2                      0.00   anomaly
3                      0.01    normal
4                      0.00    normal
```

1.4 3. Exploratory Data Analysis

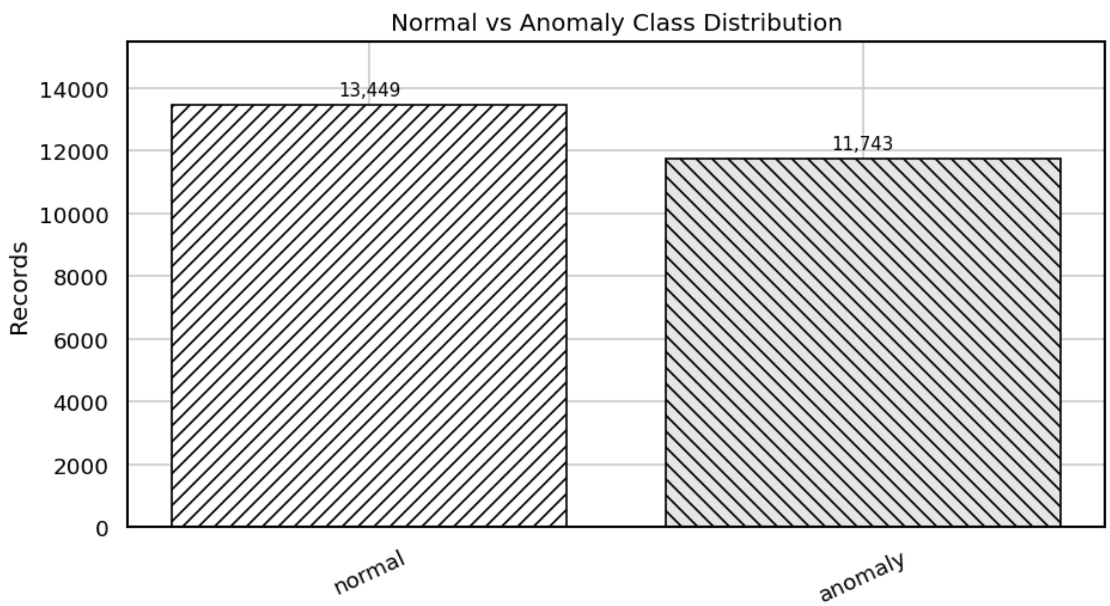
[4]:
```python
print("Columns:", df_raw.columns.tolist())
print("\nMissing values")
display(pd.DataFrame({
    "missing": df_raw.isna().sum(),
    "missing_pct": (df_raw.isna().mean() * 100).round(2),
}).sort_values("missing", ascending=False).head(15))

class_counts = df_raw["class"].value_counts()
display(class_counts.rename_axis("class").reset_index(name="count"))
plot_count_bar(class_counts, title="Normal vs Anomaly Class Distribution",␣
 ↪ylabel="Records")

for col in ["protocol_type", "flag", "service"]:
    print(f"Top values for {col}")
    display(df_raw[col].value_counts().head(12).rename_axis(col).
 ↪reset_index(name="count"))
    plot_count_bar(df_raw[col].value_counts(), title=f"Top {col} Values",␣
 ↪xlabel="Records", horizontal=True, top_n=12)
```

```
Columns: ['duration', 'protocol_type', 'service', 'flag', 'src_bytes',
'dst_bytes', 'land', 'wrong_fragment', 'urgent', 'hot', 'num_failed_logins',
'logged_in', 'num_compromised', 'root_shell', 'su_attempted', 'num_root',
'num_file_creations', 'num_shells', 'num_access_files', 'num_outbound_cmds',
'is_host_login', 'is_guest_login', 'count', 'srv_count', 'serror_rate',
'srv_serror_rate', 'rerror_rate', 'srv_rerror_rate', 'same_srv_rate',
'diff_srv_rate', 'srv_diff_host_rate', 'dst_host_count', 'dst_host_srv_count',
'dst_host_same_srv_rate', 'dst_host_diff_srv_rate',
'dst_host_same_src_port_rate', 'dst_host_srv_diff_host_rate',
'dst_host_serror_rate', 'dst_host_srv_serror_rate', 'dst_host_rerror_rate',
'dst_host_srv_rerror_rate', 'class']

Missing values
```

	missing	missing_pct
duration	0	0.0
protocol_type	0	0.0
service	0	0.0

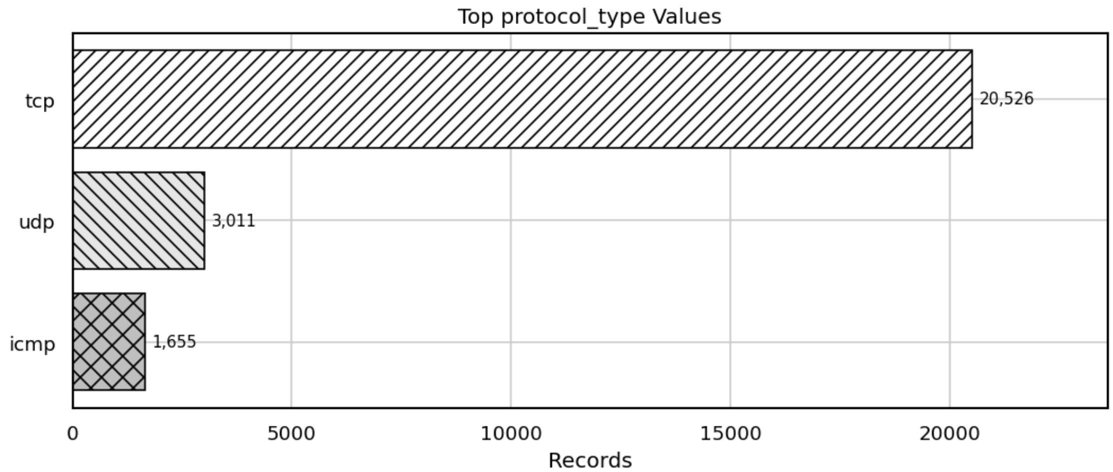
```
flag                        0          0.0
src_bytes                   0          0.0
dst_bytes                   0          0.0
land                        0          0.0
wrong_fragment              0          0.0
urgent                      0          0.0
hot                         0          0.0
num_failed_logins           0          0.0
logged_in                   0          0.0
num_compromised             0          0.0
root_shell                  0          0.0
su_attempted                0          0.0
```

```
     class  count
0   normal  13449
1  anomaly  11743
```

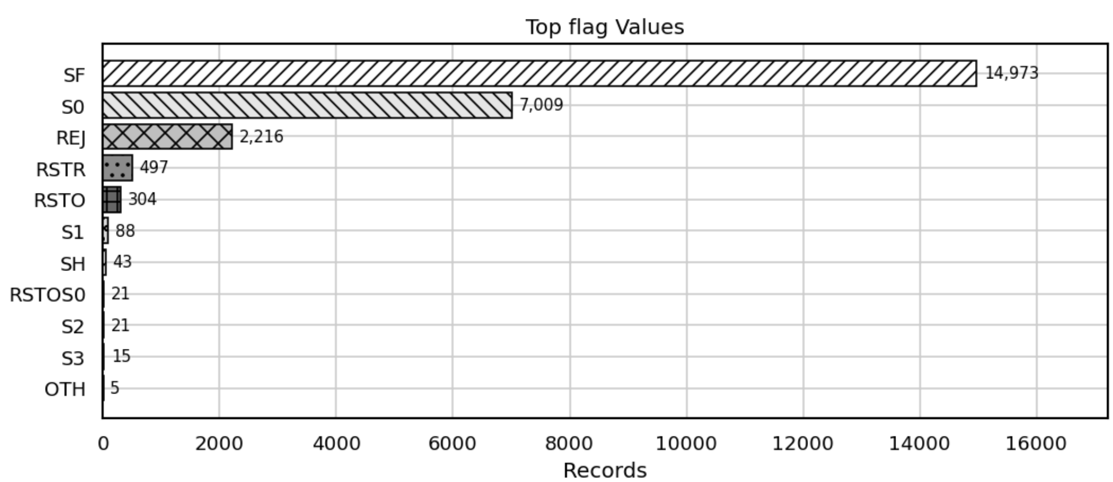



```
Top values for protocol_type
```

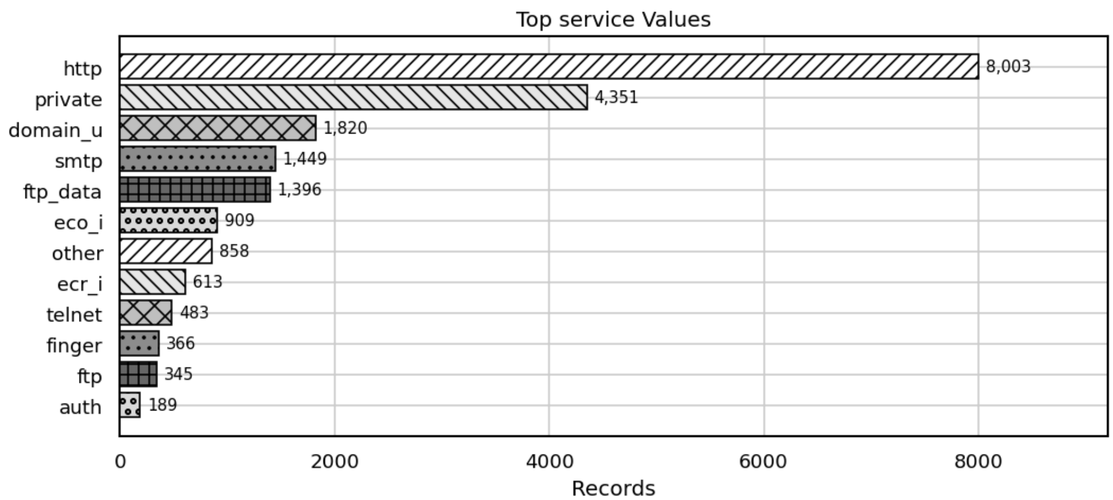
```
  protocol_type  count
0           tcp  20526
1           udp   3011
2          icmp   1655
```




```
Top values for flag

      flag  count
0       SF  14973
1       S0   7009
2      REJ   2216
3     RSTR    497
4     RSTO    304
5       S1     88
6       SH     43
7   RSTOS0     21
8       S2     21
9       S3     15
10     OTH      5
```



```
Top values for service

      service  count
0        http   8003
1     private   4351
2    domain_u   1820
3        smtp   1449
4    ftp_data   1396
5       eco_i    909
6       other    858
7       ecr_i    613
8      telnet    483
9      finger    366
10        ftp    345
11       auth    189
```



1.5 4. Sampling and Preprocessing

[5]:
```python
df = stratified_sample(df_raw, label_col="class", n_per_class=SAMPLE_PER_CLASS)
print("Sampled shape:", df.shape)
display(df["class"].value_counts().rename_axis("class").
  ↪reset_index(name="count"))

y_true = (df["class"] != "normal").astype(int).to_numpy()
X_processed, X_scaled, scaler = preprocess_features(df, label_col="class")

print("Processed feature shape:", X_processed.shape)
display(X_processed.head())
```

```
Sampled shape: (3000, 42)
```

```
     class  count
0   normal   1500
1  anomaly   1500
```

```
Processed feature shape: (3000, 109)
```

```
   duration  src_bytes  dst_bytes  land  wrong_fragment  urgent  hot  \
0         0       2444        329     0               0       0    0
1         0          0          0     0               0       0    0
2         0      10044          0     0               0       0    0
3         0       1032          0     0               0       0    0
4         0        163       1510     0               0       0    0

   num_failed_logins  logged_in  num_compromised  root_shell  su_attempted  \
0                  0          1                0           0             0
1                  0          0                0           0             0
2                  0          1                0           0             0
3                  0          0                0           0             0
4                  0          1                0           0             0

   num_root  num_file_creations  num_shells  num_access_files  \
0         0                   0           0                 0
1         0                   0           0                 0
2         0                   0           0                 0
3         0                   0           0                 0
4         0                   0           0                 0

   num_outbound_cmds  is_host_login  is_guest_login  count  srv_count  \
0                  0              0               0      2          2
1                  0              0               0    225         25
2                  0              0               0     12         12
3                  0              0               0    511        511
4                  0              0               0      3          7

   serror_rate  srv_serror_rate  rerror_rate  srv_rerror_rate  same_srv_rate  \
0          0.0              0.0          0.0              0.0           1.00
1          1.0              1.0          0.0              0.0           0.11
2          0.0              0.0          0.0              0.0           1.00
3          0.0              0.0          0.0              0.0           1.00
4          0.0              0.0          0.0              0.0           1.00

   diff_srv_rate  srv_diff_host_rate  dst_host_count  dst_host_srv_count  \
0           0.00                0.00             255                 157
1           0.05                0.00             255                  25
2           0.00                0.00              18                  18
3           0.00                0.00             255                 124
4           0.00                0.29               3                 255

   dst_host_same_srv_rate  dst_host_diff_srv_rate  \
```

```
0                   0.62                    0.02
1                   0.10                    0.05
2                   1.00                    0.00
3                   0.49                    0.02
4                   1.00                    0.00

   dst_host_same_src_port_rate  dst_host_srv_diff_host_rate  \
0                         0.00                         0.00
1                         0.00                         0.00
2                         1.00                         0.00
3                         0.49                         0.00
4                         0.33                         0.01

   dst_host_serror_rate  dst_host_srv_serror_rate  dst_host_rerror_rate  \
0                  0.00                       0.0                   0.0
1                  1.00                       1.0                   0.0
2                  0.00                       0.0                   0.0
3                  0.02                       0.0                   0.0
4                  0.00                       0.0                   0.0

   dst_host_srv_rerror_rate  protocol_type_icmp  protocol_type_tcp  \
0                       0.0               False               True
1                       0.0               False               True
2                       0.0               False               True
3                       0.0                True              False
4                       0.0               False               True

   protocol_type_udp  service_IRC  service_Z39_50  service_auth  service_bgp  \
0              False        False           False         False        False
1              False        False           False         False        False
2              False        False           False         False        False
3              False        False           False         False        False
4              False        False           False         False        False

   service_courier  service_csnet_ns  service_ctf  service_daytime  \
0            False             False        False            False
1            False             False        False            False
2            False             False        False            False
3            False             False        False            False
4            False             False        False            False

   service_discard  service_domain  service_domain_u  service_echo  \
0            False           False             False         False
1            False           False             False          True
2            False           False             False         False
3            False           False             False         False
4            False           False             False         False
```

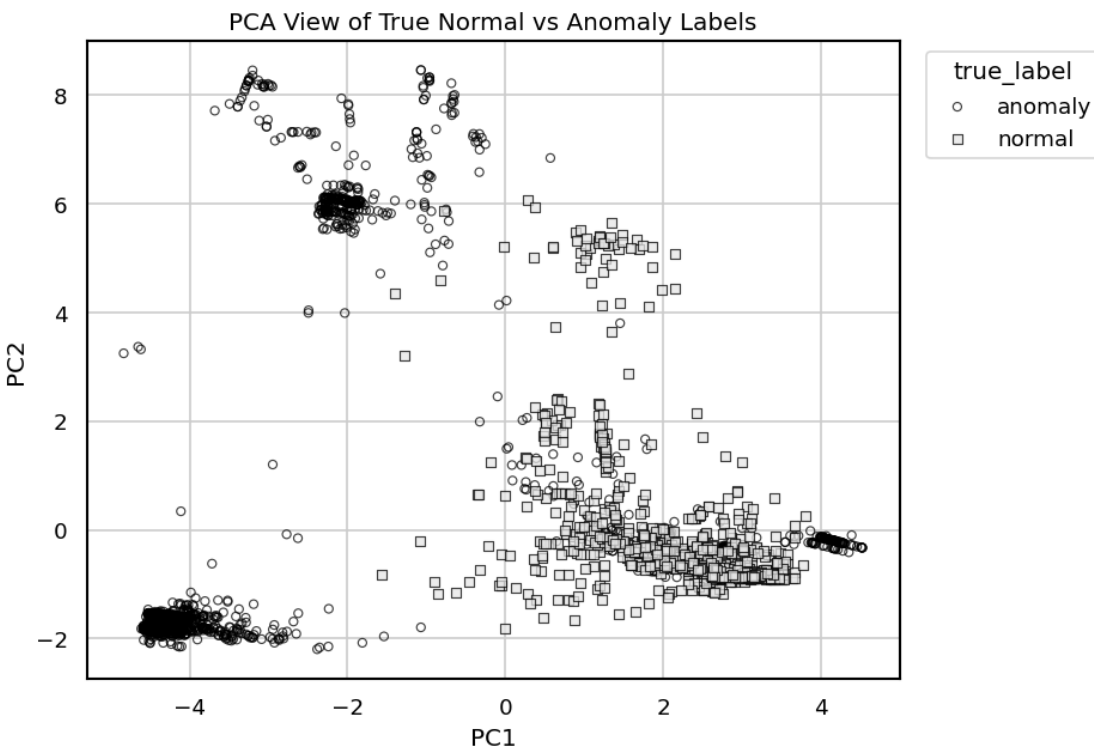
```
   service_eco_i  service_ecr_i  service_efs  service_exec  service_finger  \
0          False          False        False         False           False
1          False          False        False         False           False
2          False          False        False         False           False
3          False           True        False         False           False
4          False          False        False         False           False

   service_ftp  service_ftp_data  service_gopher  service_hostnames  \
0        False             False           False              False
1        False             False           False              False
2        False              True           False              False
3        False             False           False              False
4        False             False           False              False

   service_http  service_http_443  service_imap4  service_iso_tsap  \
0         False             False          False             False
1         False             False          False             False
2         False             False          False             False
3         False             False          False             False
4          True             False          False             False

   service_klogin  service_kshell  service_ldap  service_link  service_login  \
0           False           False         False         False          False
1           False           False         False         False          False
2           False           False         False         False          False
3           False           False         False         False          False
4           False           False         False         False          False

   service_mtp  service_name  service_netbios_dgm  service_netbios_ns  \
0        False         False                False               False
1        False         False                False               False
2        False         False                False               False
3        False         False                False               False
4        False         False                False               False

   service_netbios_ssn  service_netstat  service_nnsp  service_nntp  \
0                False            False         False         False
1                False            False         False         False
2                False            False         False         False
3                False            False         False         False
4                False            False         False         False

   service_ntp_u  service_other  service_pop_2  service_pop_3  \
0          False          False          False          False
1          False          False          False          False
2          False          False          False          False
3          False          False          False          False
4          False          False          False          False
```

```
   service_printer  service_private  service_remote_job  service_rje  \
0            False            False               False        False   
1            False            False               False        False   
2            False            False               False        False   
3            False            False               False        False   
4            False            False               False        False   

   service_shell  service_smtp  service_sql_net  service_ssh  service_sunrpc  \
0          False          True            False        False           False   
1          False         False            False        False           False   
2          False         False            False        False           False   
3          False         False            False        False           False   
4          False         False            False        False           False   

   service_supdup  service_systat  service_telnet  service_time  \
0           False           False           False         False   
1           False           False           False         False   
2           False           False           False         False   
3           False           False           False         False   
4           False           False           False         False   

   service_urp_i  service_uucp  service_uucp_path  service_vmnet  \
0          False         False              False          False   
1          False         False              False          False   
2          False         False              False          False   
3          False         False              False          False   
4          False         False              False          False   

   service_whois  flag_REJ  flag_RSTO  flag_RSTR  flag_S0  flag_S1  flag_S2  \
0          False     False      False      False    False    False    False   
1          False     False      False      False     True    False    False   
2          False     False      False      False    False    False    False   
3          False     False      False      False    False    False    False   
4          False     False      False      False    False    False    False   

   flag_SF  flag_SH  
0     True    False  
1    False    False  
2     True    False  
3     True    False  
4     True    False  
```

1.6 5. PCA Visualization

[6]:
```
pca = PCA(n_components=2, random_state=RANDOM_STATE)
X_pca = pca.fit_transform(X_scaled)
pca_df = pd.DataFrame(X_pca, columns=["PC1", "PC2"])
pca_df["true_label"] = np.where(y_true == 1, "anomaly", "normal")

print("Explained variance ratio:", np.round(pca.explained_variance_ratio_, 4))
plot_pca_scatter(pca_df, color_col="true_label", title="PCA View of True Normal␣
↪vs Anomaly Labels")
```

```
Explained variance ratio: [0.0919 0.0591]
```

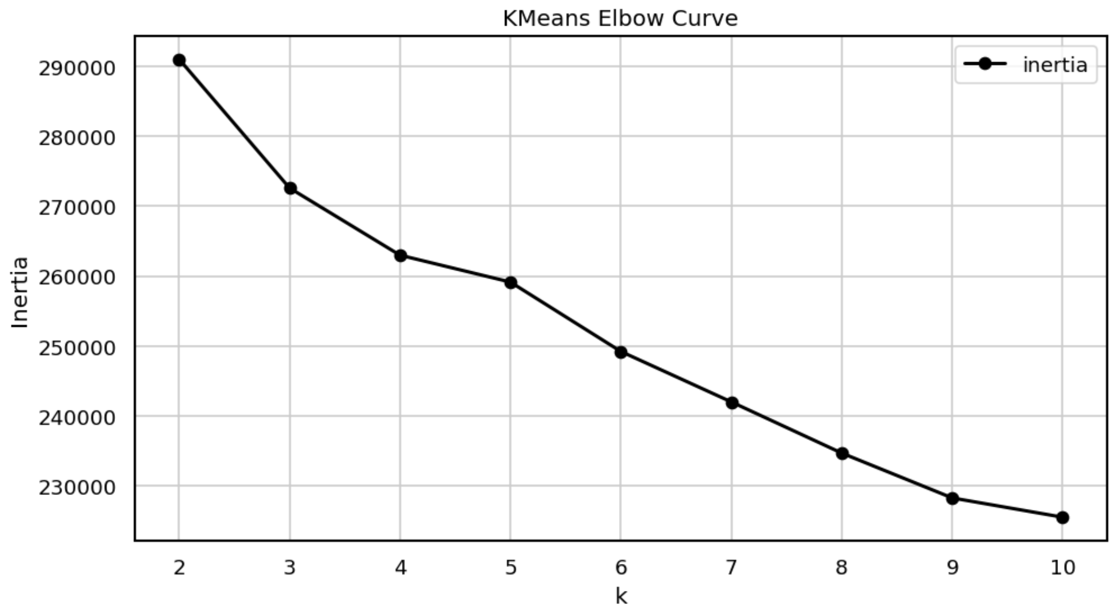


1.7 6. KMeans Clustering

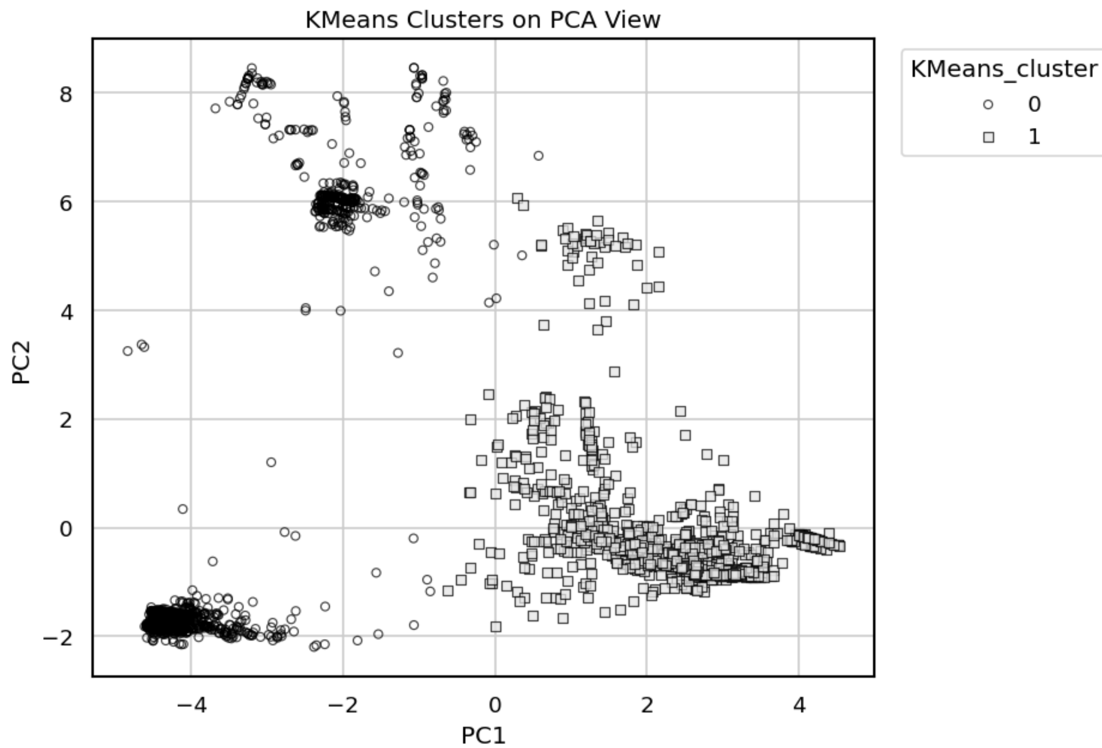
[7]:
```
k_values = range(2, 11)
kmeans_rows = []
for k in k_values:
    model = KMeans(n_clusters=k, n_init=20, random_state=RANDOM_STATE)
    labels = model.fit_predict(X_scaled)
    kmeans_rows.append({
        "k": k,
        "inertia": model.inertia_,
```

```
        "silhouette": silhouette_score(X_scaled, labels),
    })

kmeans_curve = pd.DataFrame(kmeans_rows)
display(kmeans_curve.round(4))
plot_metric_lines(kmeans_curve, "k", ["inertia"], "KMeans Elbow Curve",␣
 ↪"Inertia")
plot_metric_lines(kmeans_curve, "k", ["silhouette"], "KMeans Silhouette by k",␣
 ↪"Silhouette")

kmeans_model = KMeans(n_clusters=2, n_init=20, random_state=RANDOM_STATE)
kmeans_labels = kmeans_model.fit_predict(X_scaled)
pca_df["KMeans_cluster"] = kmeans_labels.astype(str)
plot_pca_scatter(pca_df, color_col="KMeans_cluster", title="KMeans Clusters on␣
 ↪PCA View")

results = []
predictions = {}
kmeans_result, kmeans_pred = evaluate_clustering("KMeans", kmeans_labels,␣
 ↪y_true, X_scaled)
results.append(kmeans_result)
predictions["KMeans"] = kmeans_pred
```

	k	inertia	silhouette
0	2	291026.8672	0.2203
1	3	272595.8227	0.2444
2	4	262986.8678	0.1651
3	5	259135.8047	0.1994
4	6	249211.9326	0.1817
5	7	241998.6411	0.2068
6	8	234677.1906	0.2162
7	9	228266.8111	0.2105
8	10	225509.4613	0.1957


KMeans Elbow Curve
inertia
Inertia
290000
280000
270000
260000
250000
240000
230000
2
3
4
5
6
7
8
9
10
k


KMeans Silhouette by k
silhouette
Silhouette
0.24
0.23
0.22
0.21
0.20
0.19
0.18
0.17
2
3
4
5
6
7
8
9
10
k



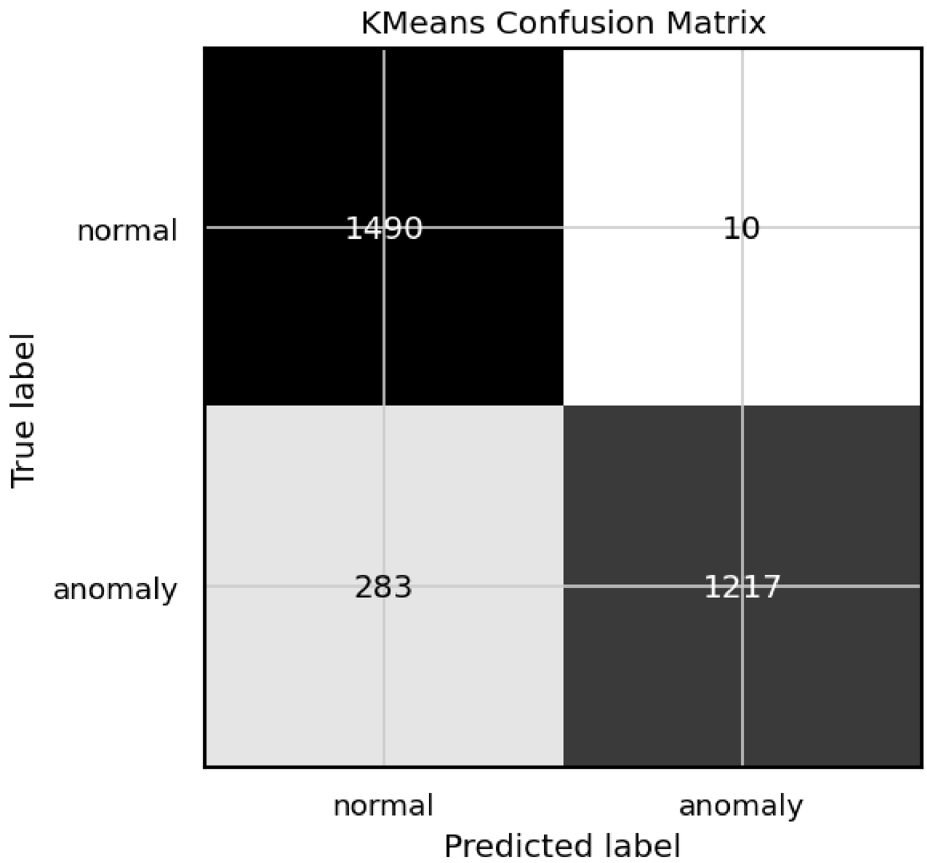
KMeans cluster-to-label mapping

	cluster	mapped_prediction	cluster_size	true_anomaly_rate	\
0	0	anomaly	1227	0.991850	
1	1	normal	1773	0.159616	

	rule
0	majority label for evaluation
1	majority label for evaluation

KMeans classification report

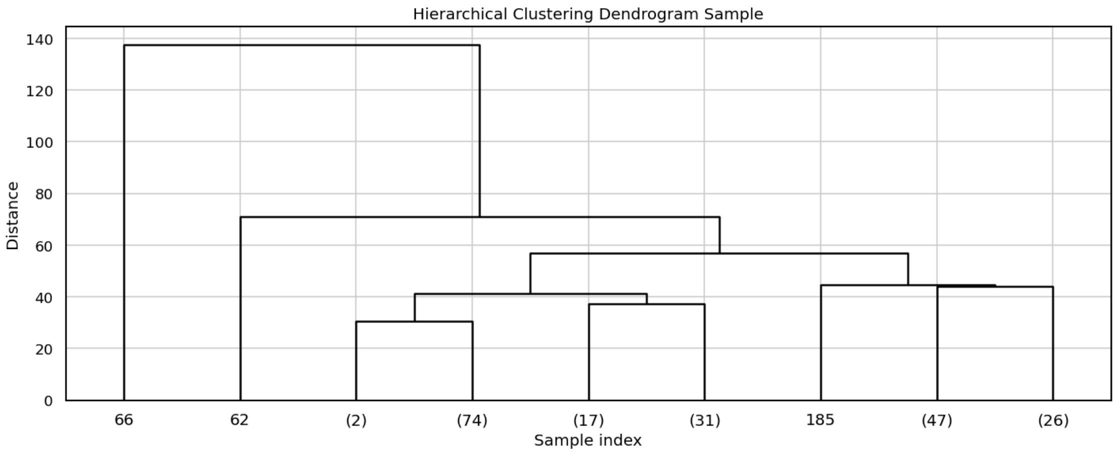
	precision	recall	f1-score	support
normal	0.8404	0.9933	0.9105	1500.0000
anomaly	0.9919	0.8113	0.8926	1500.0000
accuracy	0.9023	0.9023	0.9023	0.9023
macro avg	0.9161	0.9023	0.9015	3000.0000
weighted avg	0.9161	0.9023	0.9015	3000.0000

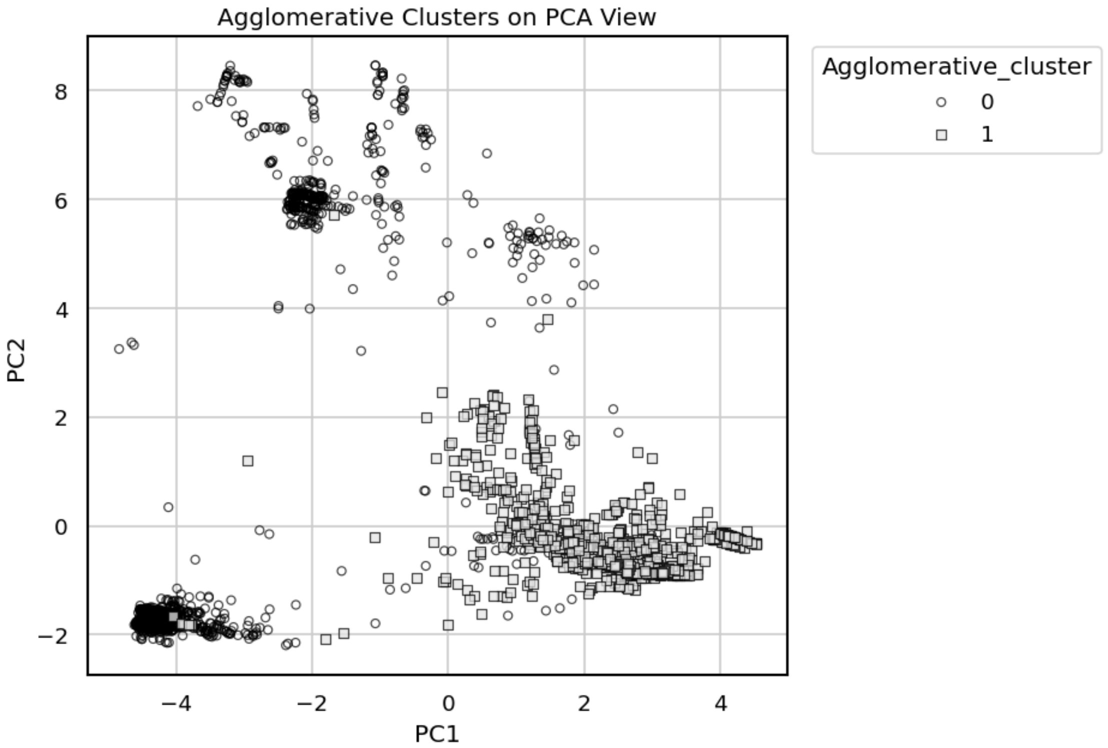


1.8 7. Agglomerative Clustering

[8]:
```
dendrogram_sample_size = min(200, X_scaled.shape[0])
rng = np.random.default_rng(RANDOM_STATE)
dendrogram_idx = rng.choice(X_scaled.shape[0], size=dendrogram_sample_size,␣
    ↪replace=False)
linked = linkage(X_scaled[dendrogram_idx], method="ward")

plt.figure(figsize=(12, 5))
dendrogram(linked, truncate_mode="level", p=4, color_threshold=0,␣
    ↪above_threshold_color="black")
plt.title("Hierarchical Clustering Dendrogram Sample")
plt.xlabel("Sample index")
plt.ylabel("Distance")
plt.tight_layout()
plt.show()

agg_model = AgglomerativeClustering(n_clusters=2, linkage="ward")
agg_labels = agg_model.fit_predict(X_scaled)
```

```python
pca_df["Agglomerative_cluster"] = agg_labels.astype(str)
plot_pca_scatter(pca_df, color_col="Agglomerative_cluster",
    title="Agglomerative Clusters on PCA View")

agg_result, agg_pred = evaluate_clustering("Agglomerative", agg_labels, y_true,
    X_scaled)
results.append(agg_result)
predictions["Agglomerative"] = agg_pred
```



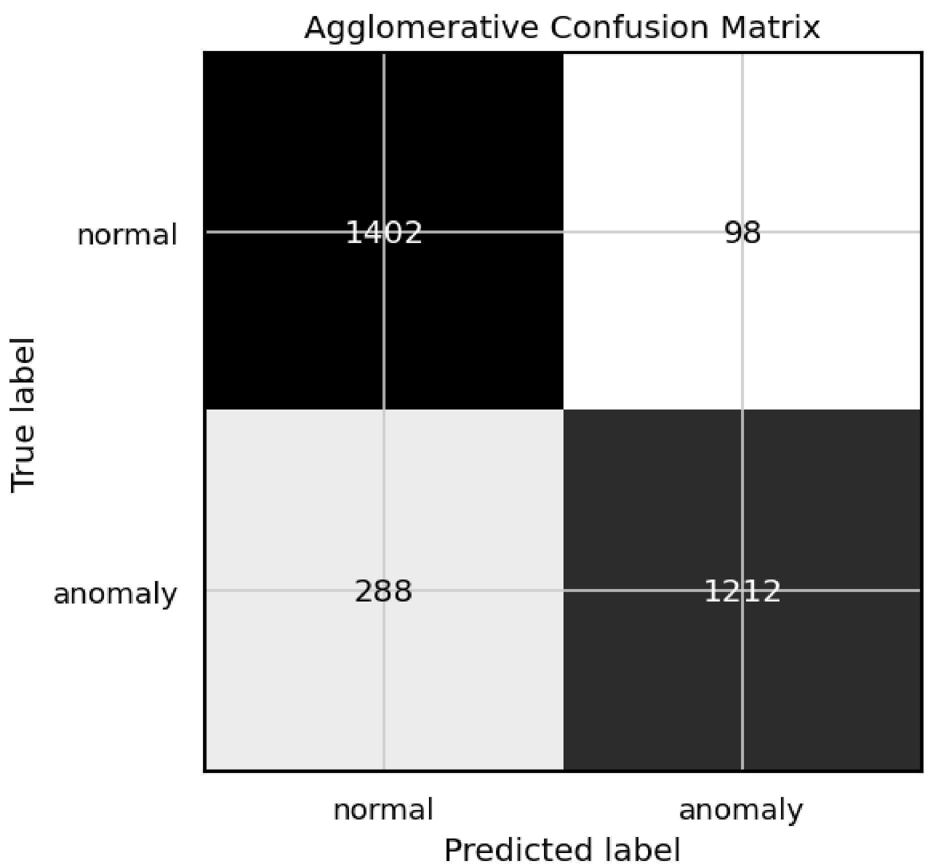


Agglomerative cluster-to-label mapping

```
   cluster mapped_prediction  cluster_size  true_anomaly_rate  \
0        0           anomaly          1310           0.925191
1        1            normal          1690           0.170414

                            rule
0  majority label for evaluation
1  majority label for evaluation
```

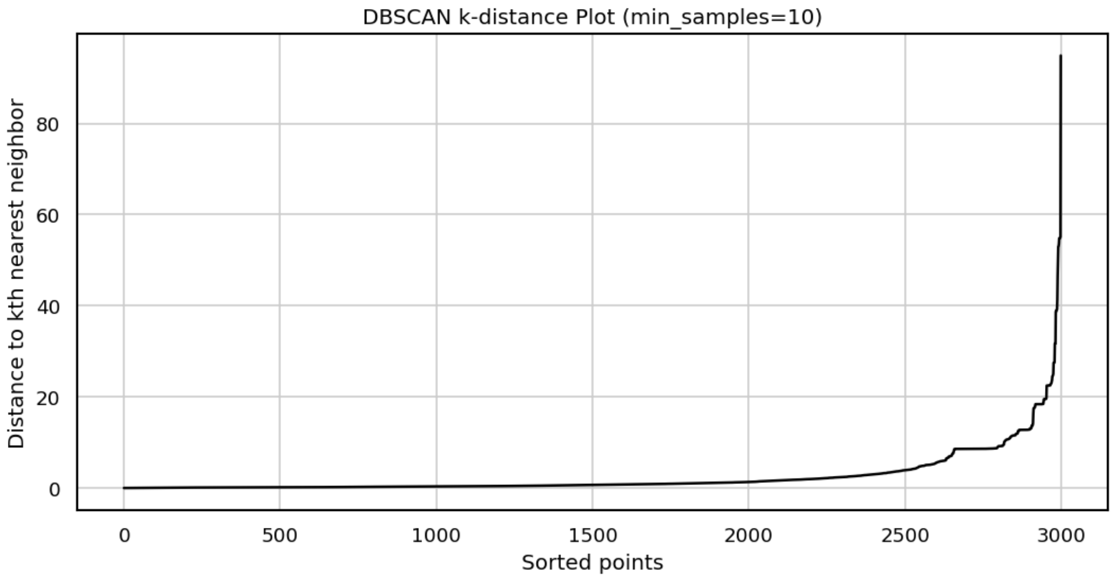
Agglomerative classification report

```
              precision  recall  f1-score    support
normal           0.8296  0.9347    0.8790  1500.0000
anomaly          0.9252  0.8080    0.8626  1500.0000
accuracy         0.8713  0.8713    0.8713     0.8713
macro avg        0.8774  0.8713    0.8708  3000.0000
weighted avg     0.8774  0.8713    0.8708  3000.0000
```



1.9 8. DBSCAN Hyperparameter Exploration

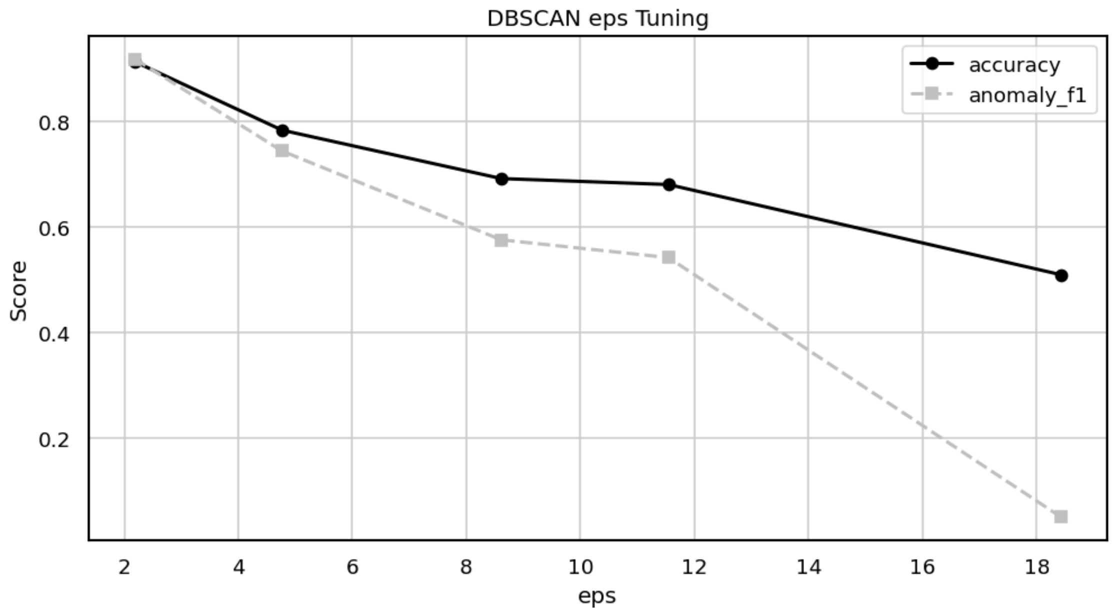
[9]:
```
min_samples = 10
neighbors = NearestNeighbors(n_neighbors=min_samples)
neighbors.fit(X_scaled)
distances, _ = neighbors.kneighbors(X_scaled)
k_distances = np.sort(distances[:, -1])

plt.figure(figsize=(9, 4.8))
plt.plot(np.arange(len(k_distances)), k_distances, color="black", linewidth=1.5)
plt.title(f"DBSCAN k-distance Plot (min_samples={min_samples})")
plt.xlabel("Sorted points")
plt.ylabel("Distance to kth nearest neighbor")
plt.tight_layout()
plt.show()

eps_candidates = np.quantile(k_distances, [0.75, 0.85, 0.90, 0.95, 0.98])
dbscan_rows = []
for eps in eps_candidates:
    labels = DBSCAN(eps=float(eps), min_samples=min_samples, n_jobs=-1).
 ↪fit_predict(X_scaled)
    pred, _ = map_clusters_to_anomaly(labels, y_true)
    dbscan_rows.append({
        "eps": float(eps),
        "clusters": len(set(labels)) - (1 if -1 in labels else 0),
        "noise_points": int((labels == -1).sum()),
        "noise_pct": (labels == -1).mean() * 100,
        "accuracy": accuracy_score(y_true, pred),
        "anomaly_f1": f1_score(y_true, pred, zero_division=0),
        **internal_cluster_metrics(X_scaled, labels, ignore_noise=True),
    })

dbscan_tuning = pd.DataFrame(dbscan_rows)
display(dbscan_tuning.round(4))
plot_metric_lines(dbscan_tuning, "eps", ["accuracy", "anomaly_f1"], "DBSCAN eps␣
 ↪Tuning", "Score")
plot_metric_lines(dbscan_tuning, "eps", ["noise_pct"], "DBSCAN Noise Percentage␣
 ↪by eps", "Noise %")
```

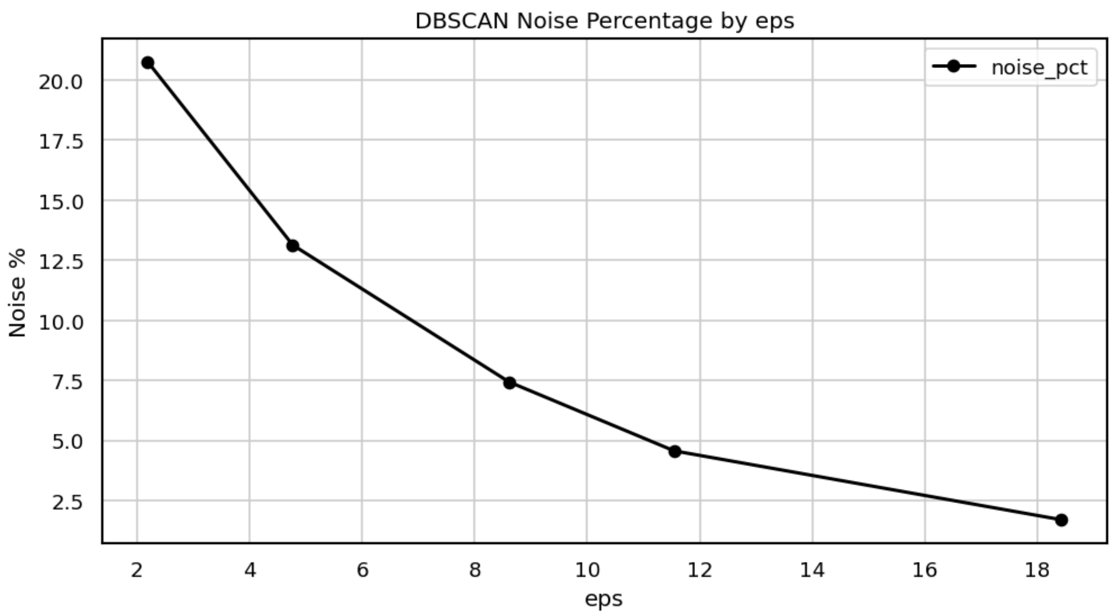


	eps	clusters	noise_points	noise_pct	accuracy	anomaly_f1
0	2.1874	48	623	20.7667	0.9123	0.9174
1	4.7672	41	394	13.1333	0.7827	0.7431
2	8.6115	31	223	7.4333	0.6910	0.5750
3	11.5436	34	137	4.5667	0.6797	0.5413
4	18.4213	1	51	1.7000	0.5090	0.0503

	Silhouette Score	Davies-Bouldin Index	Calinski-Harabasz Index
0	0.6983	0.3491	1399.8455
1	0.4420	0.5087	228.1475
2	0.4077	0.6014	97.4711
3	0.4765	0.5957	89.0824
4	NaN	NaN	NaN

	Metric Notes
0	Noise ignored
1	Noise ignored
2	Noise ignored
3	Noise ignored
4	Not enough non-noise clusters





1.10 9. DBSCAN Clustering

[ ]:
```
best_eps = dbscan_tuning.sort_values("anomaly_f1", ascending=False).
  ↪iloc[0]["eps"]
print("Selected eps from tuning table:", round(best_eps, 4))
```

```
dbscan_model = DBSCAN(eps=float(best_eps), min_samples=min_samples, n_jobs=-1)
dbscan_labels = dbscan_model.fit_predict(X_scaled)
pca_df["DBSCAN_cluster"] = dbscan_labels.astype(str)
plot_pca_scatter(pca_df, color_col="DBSCAN_cluster", title="DBSCAN Clusters and␣
↪Noise on PCA View")

dbscan_result, dbscan_pred = evaluate_clustering("DBSCAN", dbscan_labels,␣
↪y_true, X_scaled)
results.append(dbscan_result)
predictions["DBSCAN"] = dbscan_pred
```

1.11 10. Model Comparison

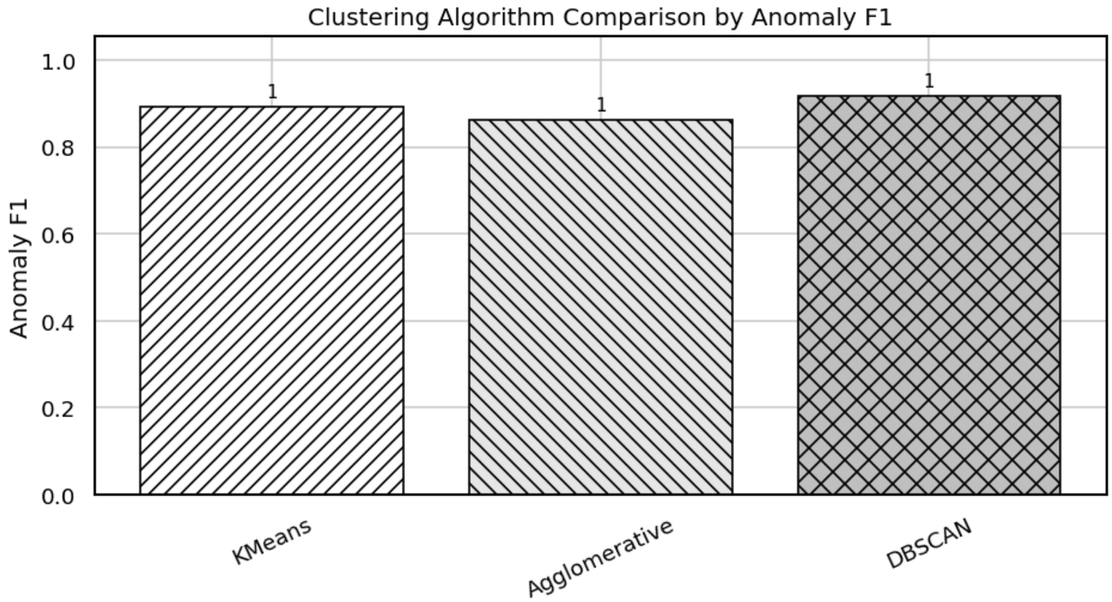
[11]:
```
comparison_df = pd.DataFrame(results)
display(comparison_df.round(4))

print("Internal clustering metrics")
internal_metrics_df = comparison_df[
    [
        "Model",
        "Clusters",
        "Noise Points",
        "Silhouette Score",
        "Davies-Bouldin Index",
        "Calinski-Harabasz Index",
        "Metric Notes",
    ]
]
display(internal_metrics_df.round(4))

print("External evaluation metrics using known labels")
external_metrics_df = comparison_df[
    [
        "Model",
        "Accuracy",
        "Weighted Precision",
        "Weighted Recall",
        "Weighted F1",
        "Anomaly F1",
    ]
]
display(external_metrics_df.round(4))

plot_count_bar(
    comparison_df.set_index("Model")["Anomaly F1"],
    title="Clustering Algorithm Comparison by Anomaly F1",
    ylabel="Anomaly F1",
```

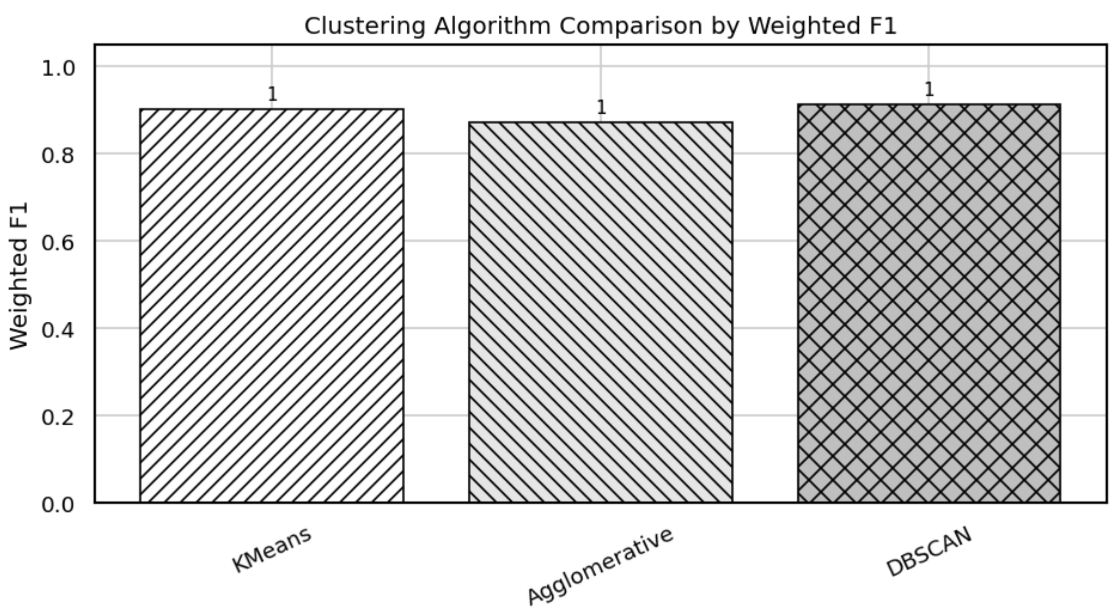
```python
)

plot_count_bar(
    comparison_df.set_index("Model")["Weighted F1"],
    title="Clustering Algorithm Comparison by Weighted F1",
    ylabel="Weighted F1",
)

plot_count_bar(
    comparison_df.set_index("Model")["Silhouette Score"],
    title="Clustering Algorithm Comparison by Silhouette Score",
    ylabel="Silhouette Score",
)

plot_count_bar(
    comparison_df.set_index("Model")["Davies-Bouldin Index"],
    title="Clustering Algorithm Comparison by Davies-Bouldin Index",
    ylabel="Davies-Bouldin Index",
)
```

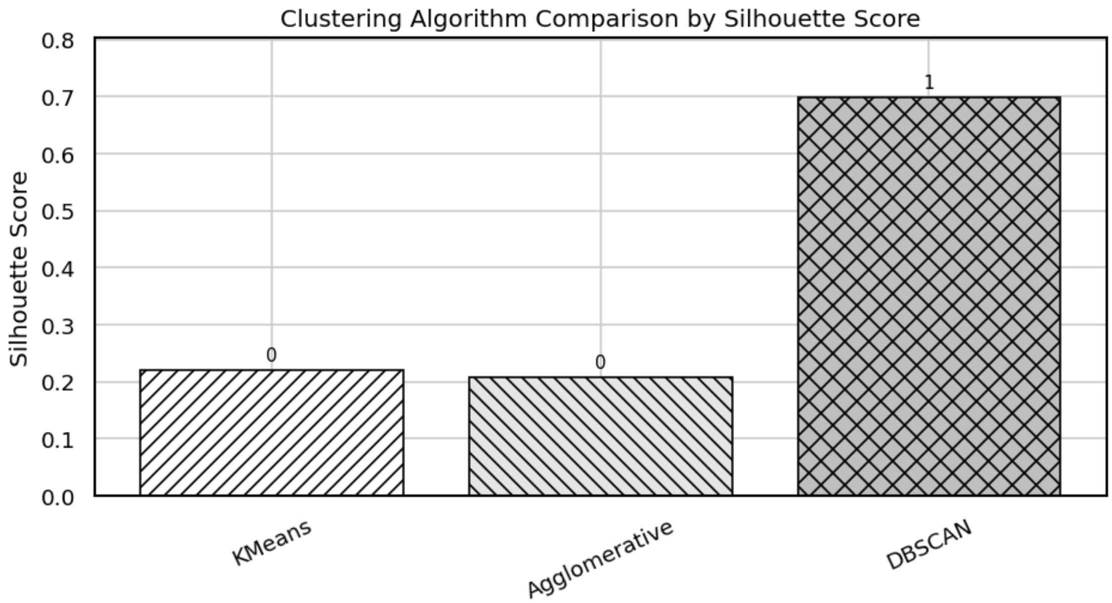
```
           Model  Clusters  Noise Points  Accuracy  Weighted Precision  \
0         KMeans         2             0    0.9023              0.9161
1  Agglomerative         2             0    0.8713              0.8774
2         DBSCAN        48           623    0.9123              0.9186

   Weighted Recall  Weighted F1  Anomaly F1  Silhouette Score  \
0           0.9023       0.9015      0.8926            0.2203
1           0.8713       0.8708      0.8626            0.2074
2           0.9123       0.9120      0.9174            0.6983

   Davies-Bouldin Index  Calinski-Harabasz Index     Metric Notes
0                2.6501                 277.8625  All points used
1                2.7249                 261.9802  All points used
2                0.3491                1399.8455    Noise ignored

Internal clustering metrics

           Model  Clusters  Noise Points  Silhouette Score  \
0         KMeans         2             0            0.2203
1  Agglomerative         2             0            0.2074
2         DBSCAN        48           623            0.6983

   Davies-Bouldin Index  Calinski-Harabasz Index     Metric Notes
0                2.6501                 277.8625  All points used
1                2.7249                 261.9802  All points used
2                0.3491                1399.8455    Noise ignored

External evaluation metrics using known labels

           Model  Accuracy  Weighted Precision  Weighted Recall  Weighted F1  \
```

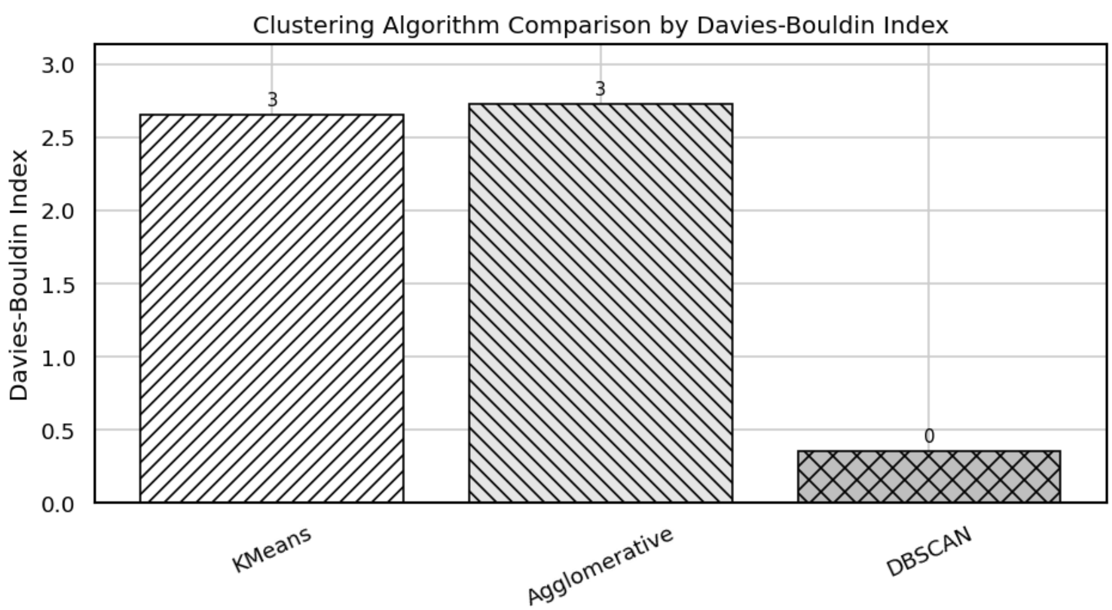
0	KMeans	0.9023	0.9161	0.9023	0.9015
1	Agglomerative	0.8713	0.8774	0.8713	0.8708
2	DBSCAN	0.9123	0.9186	0.9123	0.9120

	Anomaly F1
0	0.8926
1	0.8626
2	0.9174






Clustering Algorithm Comparison by Silhouette Score
Silhouette Score
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0
0
1
KMeans
Agglomerative
DBSCAN


Clustering Algorithm Comparison by Davies-Bouldin Index
Davies-Bouldin Index
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3
3
0
KMeans
Agglomerative
DBSCAN

1.12 11. Final Prediction Table

[12]:
```
prediction_table = pd.DataFrame({
    "actual": np.where(y_true == 1, "anomaly", "normal"),
    "KMeans": np.where(predictions["KMeans"] == 1, "anomaly", "normal"),
    "Agglomerative": np.where(predictions["Agglomerative"] == 1, "anomaly",␣
 ↪"normal"),
    "DBSCAN": np.where(predictions["DBSCAN"] == 1, "anomaly", "normal"),
})
display(prediction_table.head(20))
```

	actual	KMeans	Agglomerative	DBSCAN
0	normal	normal	normal	normal
1	anomaly	anomaly	anomaly	anomaly
2	normal	normal	normal	normal
3	anomaly	normal	normal	anomaly
4	normal	normal	normal	normal
5	anomaly	anomaly	anomaly	anomaly
6	anomaly	anomaly	anomaly	anomaly
7	normal	normal	normal	normal
8	anomaly	anomaly	anomaly	anomaly
9	normal	normal	normal	normal
10	normal	normal	normal	normal
11	anomaly	anomaly	anomaly	anomaly
12	anomaly	normal	normal	normal
13	normal	normal	normal	normal
14	normal	normal	normal	normal
15	anomaly	anomaly	anomaly	anomaly
16	anomaly	anomaly	anomaly	anomaly
17	anomaly	anomaly	anomaly	anomaly
18	normal	normal	normal	normal
19	normal	normal	normal	normal